

Operator-weighted compositions on Hardy and Bergman spaces: an exact finite-operator-span classification

Candidate substantial partial result for arXiv:1401.1697

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Status. Candidate substantial partial result, likely valid. It completely characterizes boundedness and compactness when the operator-valued weight has finite-dimensional linear span, and also characterizes weak compactness in that class for $1 < p < \infty$. The unrestricted infinite-span problem remains open.

1 Source question

Hassanlou, Laitila, and Vaezi ask for boundedness and compactness criteria for operator-valued weighted compositions on $H^p(X)$ in *Weighted composition operators between weak spaces of vector-valued analytic functions*, arXiv:1401.1697v2 [1]. Figure 1 reproduces the statement on page 6. Laitila and Tylli formulate the broader version as Question 21 of their survey, arXiv:1505.01945v1 [2]: characterize boundedness and (weak) compactness on both $H^p(X)$ and $A^p(X)$ for $1 \leq p < \infty$; see Figure 2.

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space $H_v^\infty(X)$). However, weak compactness of the more general "operator-weighted" compositions $\widetilde{W}_{\psi,\varphi}$ (where $\psi \in H(\mathbb{D}, L(X, Y))$) between $H_v^\infty(X)$ spaces has been characterized in [14].

It is worthwhile to note that the general framework (Propositions 1, 2 and 3) does not in general hold for operator-weighted composition operators. For example, an operator-weighted composition between weighted $H^\infty(X)$ spaces can be compact even if X is infinite dimensional [14]. Indeed, the present knowledge of operator-valued composition operators is quite rudimentary and, for example, their boundedness and compactness on Hardy spaces $H^p(X)$ is still an open question; see the discussion in [16, Section 7].

Figure 1: The open Hardy-space statement on page 6 of arXiv:1401.1697v2.

Question 21. Characterize boundedness and (weak) compactness of operator-weighted compositions $W_{\psi,\varphi}: H^p(X) \rightarrow H^p(Y)$ as well as $A^p(X) \rightarrow A^p(Y)$ for $1 \leq p < \infty$. The argument from [10] based on Carleson measure techniques does not readily extend to the vector-valued setting.

Figure 2: Question 21 on page 20 of arXiv:1505.01945v1.

2 Setting and a vector-lifting lemma

Let X, Y be complex Banach spaces, let $1 \leq p < \infty$, let $\varphi: \mathbb{D} \rightarrow \mathbb{D}$ be analytic, and let $\psi: \mathbb{D} \rightarrow \mathcal{L}(X, Y)$ be analytic. On either

$$\mathcal{F}_p(X) = H^p(X) \quad \text{or} \quad \mathcal{F}_p(X) = A^p(X),$$

put

$$(W_{\psi,\varphi}f)(z) = \psi(z)f(\varphi(z)). \quad (2.1)$$

For scalar analytic a , write $W_{a,\varphi}$ for the scalar weighted composition and $\widetilde{W}_{a,\varphi}^X$ for its pointwise lift to $\mathcal{F}_p(X)$. If $U: X \rightarrow Y$ is bounded, write $\widetilde{U}f(z) = Uf(z)$.

We use the following standard scalar-to-vector consequence of the Carleson embedding proof.

Lemma 1 (scalar lifting and compact postcomposition). *For $\mathcal{F}_p = H^p$ or A^p :*

- (a) *If $W_{a,\varphi}$ is bounded on scalar $\mathcal{F}_p$, then $\widetilde{W}_{a,\varphi}^X$ is bounded on $\mathcal{F}_p(X)$ for every Banach space X .*
- (b) *If in addition $W_{a,\varphi}$ is compact and $U: X \rightarrow Y$ is compact, then*

$$\widetilde{U} \widetilde{W}_{a,\varphi}^X: \mathcal{F}_p(X) \longrightarrow \mathcal{F}_p(Y) \quad (2.2)$$

is compact. No approximation property of X or Y is required.

Proof. For (a), scalar boundedness is equivalent to the appropriate Carleson embedding estimate for the pullback measure defined by a and φ . The proof of that estimate uses only the submean inequality. Apply the same proof to the nonnegative subharmonic function $z \mapsto \|f(z)\|_X^p$. This gives the vector-valued estimate, for both the boundary Carleson measure in the Hardy case and the area Carleson measure in the Bergman case.

For (b), let $K = \overline{U^*(B_{Y^*})}$, which is norm compact. Given $\delta > 0$, choose $x_1^*, \dots, x_m^* \in K$ forming a δ -net. For $h \in \mathcal{F}_p(X)$, pointwise duality gives

$$\begin{aligned} \|U(a(z)h(\varphi(z)))\|_Y &\leq \max_{1 \leq j \leq m} |a(z)x_j^*(h(\varphi(z)))| \\ &\quad + \delta |a(z)| \|h(\varphi(z))\|_X. \end{aligned} \quad (2.3)$$

Each map $h \mapsto W_{a,\varphi}(x_j^* \circ h)$ from $\mathcal{F}_p(X)$ to scalar $\mathcal{F}_p$ is compact. The finite product of these maps sends the unit ball to a totally bounded set. Equation (2.3), the lattice inequality $\|\max_j |g_j|\|_p \leq (\sum_j \|g_j\|_p^p)^{1/p}$, and the uniform bound from (a) show that sufficiently close points in that finite product have images under (2.2) within $O(\delta)$. Hence (2.2) maps the unit ball to a totally bounded set and is compact. $\square$

3 Proof intuition

Finite-dimensional operator range eliminates the genuinely operator-valued Carleson obstruction. Choose a basis $U_1, \dots, U_N$ of the linear span of $\psi(\mathbb{D})$ and write

$$\psi(z) = \sum_{k=1}^N a_k(z) U_k. \quad (3.1)$$

The forward reduction is immediate: the operator is the finite sum of the vector lifts of the scalar weighted compositions, followed by U_k .

The reverse reduction is the key point. On the finite-dimensional operator space $E = \text{span } \psi(\mathbb{D})$, the elementary functionals $U \mapsto y^*(Ux)$ span E^* . Consequently each coefficient a_k is a finite linear combination of scalarizations $y^*\psi(\cdot)x$, and the corresponding scalar weighted composition factors through the original operator. Compactness also forces each value $\psi(z)$ to be compact by testing constant functions and evaluating at z . Minimality of (3.1) provides finitely many values $\psi(z_j)$ spanning E , so this forces every U_k to be compact. Weak compactness is handled identically and its converse uses factorization of weakly compact operators through reflexive spaces.

4 The finite-span classification

Theorem 2 (boundedness, compactness, and weak compactness). *Assume $\psi \not\equiv 0$ and*

$$E = \text{span}\{\psi(z) : z \in \mathbb{D}\}$$

has finite dimension N . Choose a basis $U_1, \dots, U_N$ of E and the unique analytic coefficient functions $a_1, \dots, a_N$ in (3.1). Then, for $\mathcal{F}_p = H^p$ or A^p :

- (i) $W_{\psi, \varphi} : \mathcal{F}_p(X) \rightarrow \mathcal{F}_p(Y)$ *is bounded if and only if every scalar $W_{a_k, \varphi} : \mathcal{F}_p \rightarrow \mathcal{F}_p$ is bounded.*
- (ii) $W_{\psi, \varphi}$ *is compact if and only if every $U_k : X \rightarrow Y$ and every scalar $W_{a_k, \varphi}$ is compact.*
- (iii) *If $1 < p < \infty$ and $W_{\psi, \varphi}$ is bounded, then $W_{\psi, \varphi}$ is weakly compact if and only if every $U_k : X \rightarrow Y$ is weakly compact.*

The conditions do not depend on the chosen basis of E .

Proof. Every linear functional on E is a finite linear combination of elementary functionals $U \mapsto y^*(Ux)$. Indeed, those elementary functionals separate points of E , so their restrictions span E^* by finite dimensionality. In particular, for each coordinate functional λ_k of the basis (U_k) , there are finitely many $x_{k,l} \in X$, $y_{k,l}^* \in Y^*$, and $c_{k,l} \in \mathbb{C}$ such that

$$a_k(z) = \lambda_k(\psi(z)) = \sum_l c_{k,l} y_{k,l}^*(\psi(z)x_{k,l}). \quad (4.1)$$

Let $J_x : g \mapsto gx$ and $Q_{y^*} : f \mapsto y^* \circ f$. Then

$$W_{a_k, \varphi} = \sum_l c_{k,l} Q_{y_{k,l}^*} W_{\psi, \varphi} J_{x_{k,l}}. \quad (4.2)$$

Thus boundedness, compactness, or weak compactness of $W_{\psi, \varphi}$ passes to every scalar coefficient operator.

Conversely, (3.1) gives the operator identity

$$W_{\psi,\varphi} = \sum_{k=1}^N \tilde{U}_k \tilde{W}_{a_k,\varphi}^X. \quad (4.3)$$

Lemma 1(a) proves the converse in (i). If all U_k and all $W_{a_k,\varphi}$ are compact, Lemma 1(b) makes every summand in (4.3) compact, proving one direction of (ii).

For the reverse direction in (ii), let $j : X \rightarrow \mathcal{F}_p(X)$ send x to the constant function x , and let $\delta_z : \mathcal{F}_p(Y) \rightarrow Y$ be evaluation at z . Then

$$\psi(z) = \delta_z W_{\psi,\varphi} j. \quad (4.4)$$

Hence compactness of $W_{\psi,\varphi}$ makes every $\psi(z)$ compact. Since (3.1) is minimal, the vectors $(a_1(z), \dots, a_N(z))$ span $\mathbb{C}^N$ as z varies. Choose $z_1, \dots, z_N$ for which the coefficient matrix is invertible. Each U_k is then a linear combination of $\psi(z_1), \dots, \psi(z_N)$ and is compact. Equation (4.2) already showed that every scalar $W_{a_k,\varphi}$ is compact.

The same evaluation argument shows that weak compactness of $W_{\psi,\varphi}$ forces every U_k to be weakly compact. Conversely, let $1 < p < \infty$ and factor a weakly compact U_k as

$$X \xrightarrow{A_k} Z_k \xrightarrow{B_k} Y \quad (4.5)$$

through a reflexive Banach space Z_k , using the Davis–Figiel–Johnson–Pełczyński factorization theorem [3]. The space $\mathcal{F}_p(Z_k)$ is reflexive, and

$$\tilde{U}_k \tilde{W}_{a_k,\varphi}^X = \tilde{B}_k \tilde{W}_{a_k,\varphi}^{Z_k} \tilde{A}_k \quad (4.6)$$

factors through it. Each summand in (4.3) is therefore weakly compact, which proves (iii). Finally, a change of basis makes finite linear combinations on both the operator and scalar sides, so the stated conditions are intrinsic. $\square$

Corollary 3 (constant operator weights, including $p = 1$). *Let $0 \neq U \in \mathcal{L}(X, Y)$ and put $\psi(z) = U$. On either H^p or A^p :*

- (a) $W_{U,\varphi}$ is compact if and only if U and scalar C_φ are compact.
- (b) If $1 < p < \infty$, $W_{U,\varphi}$ is weakly compact if and only if U is weakly compact.
- (c) If $p = 1$, $W_{U,\varphi}$ is weakly compact if and only if U is weakly compact and scalar C_φ is compact.

Proof. Only (c) is not contained in Theorem 2. Necessity follows by factoring U through constant functions and C_φ through $W_{U,\varphi}$ using one $x \in X$ and $y^* \in Y^*$ with $y^*(Ux) = 1$. Weak compactness of scalar C_φ on H^1 implies compactness by Sarason’s theorem; on A^1 , it does so because A^1 has the Schur property. Conversely, factor weakly compact U through a reflexive Z . For reflexive Z , compactness of scalar C_φ implies weak compactness of the vector composition on $H^1(Z)$ and on $A^1(Z)$ by the vector-valued composition theorems summarized in [2, Section 3]. The factorization (4.6) with $a = 1$ finishes the proof. $\square$

5 Scope, verification, and novelty

The theorem gives a complete answer whenever $\psi(\mathbb{D})$ has finite linear span, not merely when X or Y is finite dimensional. It includes nonconstant operator weights and does not assume approximation properties. It does not settle the infinite-span case. There, the natural approximation route would require uniform finite-span approximation of ψ in a tensor norm that controls the vector Carleson embedding; pointwise norm approximation alone is insufficient. At $p = 1$, the weak-compactness statement for general nonconstant finite-span weights is also left open because the required weighted reflexive-target lifting is not part of the cited unweighted theorem.

The proof was audited at four possible cancellation points: elementary functionals really span E^* ; minimality supplies an invertible evaluation matrix; (2.3) uses compactness of $U^*(B_{Y^*})$ and therefore does not assume finite-rank approximation; and (4.6) has the correct order of pointwise maps. Eight focused upgrade routes are recorded in the accompanying attempt note.

Repository indexes and bounded exact-phrases, arXiv/title/author, constant operator-weight, finite-operator-span, and later-work searches through 13 August 2026 found the scalar theory and the source survey but no explicit statement of Theorem 2. Novelty is therefore provisional, not a claim of priority. Human review should focus first on the scalar-to-vector Carleson lifting in Lemma 1(a), and then on the compactness finite-net estimate (2.3).

References

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- [2] J. Laitila and H.-O. Tylli, *Composition operators on vector-valued analytic function spaces: a survey*, arXiv:1505.01945v1; Acta Comment. Univ. Tartu. Math. 18 (2014), 135–157.
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